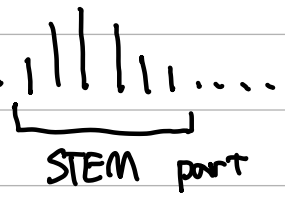
Chapter 05

- $x[n] \rightarrow$ FIR filter $\rightarrow y[n]$

- 

- 3 point average system

$$y[n] = h[n] * x[n]$$

convolution

$$= \sum_{k=-\infty}^{\infty} h[k] \cdot x[n-k]$$

$$h[n] * x[n] = x[n] * h[n]$$

- convolution sum

$y[n] = h[n] * x[n]$ 증명

$$x[n] = \sum_{l} x[l]\,\delta[n-l]$$

$$= \cdots + x[0]\,\delta[n] + x[1]\,\delta[n-1] + \cdots + x[l]\,\delta[n-l]$$

$\delta[n] \mapsto h[n]$

$\delta[n-1] \mapsto h[n-1]$

$\vdots$

$\delta[n-l] \mapsto h[n-l]$

} time invariance

$x[0]\,\delta[n] \mapsto x[0]\,h[n]$

$\vdots$

$x[l]\,\delta[n-l] \mapsto x[l]\,h[n-l]$

} scaling

$$x[n] = \sum_{l=-\infty}^{\infty} x[l]\,\delta[n-l] \mapsto \sum_{l=-\infty}^{\infty} x[l]\,h[n-l] = x[n] * h[n]$$

- Commutative property of convolution

$$y[n] = x[n] * h[n] = h[n] * x[n]$$

$k = n-l \Rightarrow l = n-k$

$$x[n] * h[n] = \sum_{l=-\infty}^{\infty} x[l]\,h[n-l]$$

$$= \sum_{k=+\infty}^{-\infty} x[n-k]\,h[k] = \sum_{k=\infty}^{-\infty} h[k]\,x[n-k]$$



교환기 순서 변경

$$= \sum_{k=-\infty}^{\infty} h[k]\,x[n-k] = h[n] * x[n]$$

더하기 순서 변경